# PolynomialFeatures

*class* sklearn.preprocessing.**PolynomialFeatures**(*degree=2, *, interaction_only=False, include_bias=True, order='C'*) [source]

Generate polynomial and interaction features.

Generate a new feature matrix consisting of all polynomial combinations of the features with degree less than or equal to the specified degree. For example, if an input sample is two dimensional and of the form [a, b], the degree-2 polynomial features are [1, a, b, a^2, ab, b^2].

Read more in the User Guide.

**Parameters:**

**degree** : ***int or tuple (min_degree, max_degree), default=2***

If a single int is given, it specifies the maximal degree of the polynomial features. If a tuple `(min_degree, max_degree)` is passed, then `min_degree` is the minimum and `max_degree` is the maximum polynomial degree of the generated features. Note that `min_degree=0` and `min_degree=1` are equivalent as outputting the degree zero term is determined by `include_bias`.

**interaction_only** : ***bool, default=False***

If `True`, only interaction features are produced: features that are products of at most `degree` *distinct* input features, i.e. terms with power of 2 or higher of the same input feature are excluded:

- included: `x[0]`, `x[1]`, `x[0] * x[1]`, etc.
- excluded: `x[0] ** 2`, `x[0] ** 2 * x[1]`, etc.

**include_bias** : ***bool, default=True***

If `True` (default), then include a bias column, the feature in which all polynomial powers are zero (i.e. a column of ones - acts as an intercept term in a linear model).

**order** : ***{'C', 'F'}, default='C'***

Order of output array in the dense case. `'F'` order is faster to compute, but may slow down subsequent estimators.

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*Added in version 0.21.*

**Attributes:**

`powers_` : *ndarray of shape* ( `n_output_features_` , `n_features_in_` )

Exponent for each of the inputs in the output.

**n_features_in_** : *int*

Number of features seen during fit.

> *Added in version 0.24.*

**feature_names_in_** : *ndarray of shape* ( `n_features_in_` ,)

Names of features seen during fit. Defined only when `X` has feature names that are all strings.

> *Added in version 1.0.*

**n_output_features_** : *int*

The total number of polynomial output features. The number of output features is computed by iterating over all suitably sized combinations of input features.

See also

`SplineTransformer`

Transformer that generates univariate B-spline bases for features.

## Notes

Be aware that the number of features in the output array scales polynomially in the number of features of the input array, and exponentially in the degree. High degrees can cause overfitting.

See examples/linear_model/plot_polynomial_interpolation.py

## Examples

```
>>> import numpy as np
>>> from sklearn.preprocessing import PolynomialFeatures
>>> X = np.arange(6).reshape(3, 2)
>>> X
array([[0, 1],
       [2, 3],
       [4, 5]])
>>> poly = PolynomialFeatures(2)
>>> poly.fit_transform(X)
array([[ 1.,  0.,  1.,  0.,  0.,  1.],
       [ 1.,  2.,  3.,  4.,  6.,  9.],
       [ 1.,  4.,  5., 16., 20., 25.]])
>>> poly = PolynomialFeatures(interaction_only=True)
>>> poly.fit_transform(X)
array([[ 1.,  0.,  1.,  0.],
       [ 1.,  2.,  3.,  6.],
       [ 1.,  4.,  5., 20.]])
```

### `fit(X, y=None)` [source]

Compute number of output features.

**Parameters:**

**X** : ***{array-like, sparse matrix} of shape (n_samples, n_features)***

The data.

**y** : ***Ignored***

Not used, present here for API consistency by convention.

**Returns:**

**self** : ***object***

Fitted transformer.

### `fit_transform(X, y=None, **fit_params)` [source]

Fit to data, then transform it.

Fits transformer to `X` and `y` with optional parameters `fit_params` and returns a transformed version of `X`.

**Parameters:**

**X** : ***array-like of shape (n_samples, n_features)***

Input samples.

**y** : ***array-like of shape (n_samples,) or (n_samples, n_outputs), default=None***

Target values (None for unsupervised transformations).

****fit_params** : ***dict***

Additional fit parameters. Pass only if the estimator accepts additional params in its `fit` method.

**Returns:**

**X_new** : ***ndarray array of shape (n_samples, n_features_new)***

Transformed array.

## `get_feature_names_out(input_features=None)` [source]

Get output feature names for transformation.

**Parameters:**

**input_features** : ***array-like of str or None, default=None***

Input features.

- If `input_features is None`, then `feature_names_in_` is used as feature names in. If `feature_names_in_` is not defined, then the following input feature names are generated: `["x0", "x1", ..., "x(n_features_in_ - 1)"]`.
- If `input_features` is an array-like, then `input_features` must match `feature_names_in_` if `feature_names_in_` is defined.

**Returns:**

**feature_names_out** : ***ndarray of str objects***

Transformed feature names.

## `get_metadata_routing()` [source]

Get metadata routing of this object.

Please check User Guide on how the routing mechanism works.

**Returns:**

**routing** : ***MetadataRequest***

A `MetadataRequest` encapsulating routing information.

`get_params(deep=True)` [source]

Get parameters for this estimator.

**Parameters:**

**deep** : ***bool, default=True***

If True, will return the parameters for this estimator and contained subobjects that are estimators.

**Returns:**

**params** : ***dict***

Parameter names mapped to their values.

`set_output(*, transform=None)` [source]

Set output container.

See Introducing the set_output API for an example on how to use the API.

**Parameters:**

**transform** : ***{"default", "pandas", "polars"}, default=None***

Configure output of `transform` and `fit_transform`.

- `"default"` : Default output format of a transformer
- `"pandas"` : DataFrame output
- `"polars"` : Polars output
- `None` : Transform configuration is unchanged

> ***Added in version 1.4:*** `"polars"` option was added.

**Returns:**

**self** : ***estimator instance***

Estimator instance.

`set_params(**params)` [source]

Set the parameters of this estimator.

The method works on simple estimators as well as on nested objects (such as `Pipeline`). The latter have parameters of the form `<component>__<parameter>` so that it's possible to update each component of a nested object.

**Parameters:**

****params** : *dict*

Estimator parameters.

**Returns:**

**self** : *estimator instance*

Estimator instance.

`transform(X)` [source]

Transform data to polynomial features.

**Parameters:**

**X** : *{array-like, sparse matrix} of shape (n_samples, n_features)*

The data to transform, row by row.

Prefer CSR over CSC for sparse input (for speed), but CSC is required if the degree is 4 o higher. If the degree is less than 4 and the input format is CSC, it will be converted to CSR, have its polynomial features generated, then converted back to CSC.

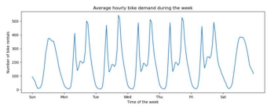
If the degree is 2 or 3, the method described in "Leveraging Sparsity to Speed Up Polynomial Feature Expansions of CSR Matrices Using K-Simplex Numbers" by Andrew Nystrom and John Hughes is used, which is much faster than the method used on CSC input. For this reason, a CSC input will be converted to CSR, and the output will be converted back to CSC prior to being returned, hence the preference of CSR.

**Returns:**

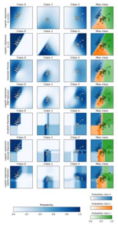
**XP** : *{ndarray, sparse matrix} of shape (n_samples, NP)*

The matrix of features, where `NP` is the number of polynomial features generated from the combination of inputs. If a sparse matrix is provided, it will be converted into a sparse `csr_matrix`.

# Gallery examples

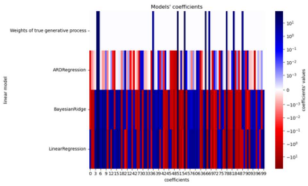


Time-related feature engineering

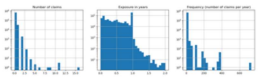


Plot classification probability

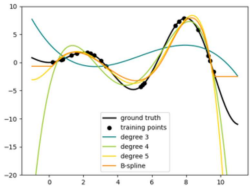
Visualizing the probabilistic predictions of a VotingClassifier



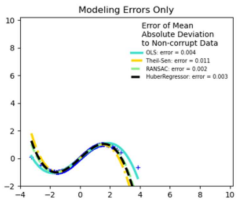
Comparing Linear Bayesian Regressors



Poisson regression and non-normal loss



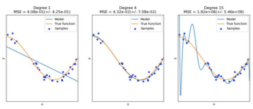

Polynomial and Spline interpolation

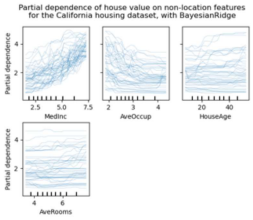


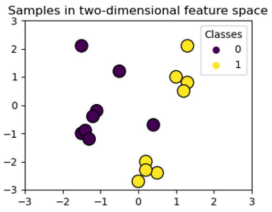

Robust linear estimator fitting



Displaying Pipelines



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